

CS & IT ENGINEERING

Discrete Maths
Set Theory



Lecture No.1



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TOPICS TO BE
COVERED

01 Basics of sets

02 Types of sets

03 Types of Interval

04 Set Operations

05 Set formulas

Set: unordered collections of an objects.

A : a, b, c object/element/member(a) $a \in A$
 $\downarrow$
 element $\in$ Set.

$$A = \left\{ \underbrace{n}_{e_1}, \underbrace{\{n\}}_{e_2}, \underbrace{\{\{n\}\}}_{e_3} \right\}$$



1. order does not matter.

$\{a, b, c\}$ or $\{b, a, c\}$

element $\in$ Set

e_2

$$\{n\} \in A$$

e_3

$$\{\{n\}\} \in A$$



2. Reptⁿ are not allowed.

Roster method:

$$\{1, 2, 3\}$$

$$\{1, \dots, 100\}$$

$$\{1, \dots, \infty\}$$

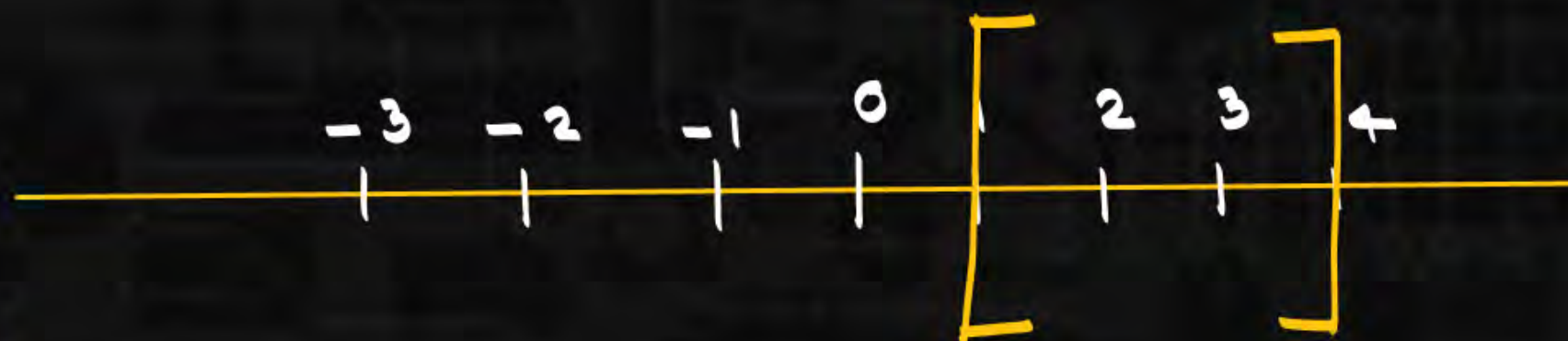
Set builder notation:

$$\{ \text{element} \mid \text{conditions} \}$$

$$\{x \mid 1 \leq x \leq 3\}$$

$$\{x \mid x \text{ is perfect square}\}$$

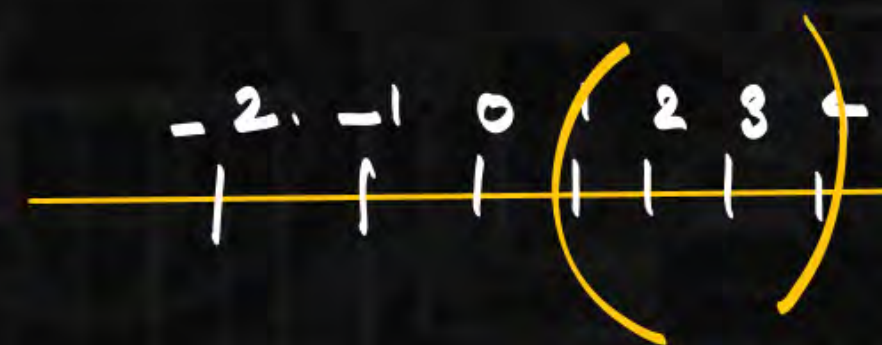
Interval:



$$\underline{[1, 4]} = \{1, 2, 3, 4\}$$

$$[a, b] = \{x \mid a \leq x \leq b\}$$

$$(1, 7) = \{2, 3, 4, 5, 6\}$$



$$(a, b) = \{x \mid a < x < b\}$$

$$[1, 7) = \{1, 2, 3, 4, 5, 6\}$$

$$A_n = [-2n, 3n]$$

$$A_1 = [-2, 3]$$

$$A_2 = [-4, 6]$$



equal sets:

A, B

$$\forall x (x \in A \leftrightarrow x \in B) \quad A = B$$

Cardinality: $|A|$ = no. of elements present in set

empty set: $(\emptyset)$

$$A = \boxed{} \quad A = \{ \}$$

$$|A| = 0$$

eg2: $A = \{ \{ \} \}$ or $\{ \emptyset \}$

$$|A| = 1$$

$|A| = 1$ (singleton set)



{ }

Subset ($\subseteq$)

set 1 $\subseteq$ set 2.

$$A = \{1, 2\} \quad B = \{1, 2, 3\}$$

$$A \subseteq B$$

$$A \subseteq B$$

Any set S.

$$\forall x (x \in A \rightarrow x \in B)$$

$$\begin{array}{l} 1) \quad \underline{\emptyset \subseteq S.} \\ 2) \quad S \subseteq S. \end{array}$$

$$\forall x (x \in \emptyset \rightarrow x \in S)$$

proper subset ($\subset$)

$$A \subset B$$

$$|A| < |B|$$

$$\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$$

$\subset$

$$A \subseteq B$$

$\subseteq$

$$|A| \leq |B|$$

↓

$$|A| < |B| \text{ OR } |A| = |B|$$

$\subset$ OR $=$

↓

$\subseteq$

Set: A

Power set: $(P(A) / \underline{2^A}) \longrightarrow$ set of all subsets.

$$A = \{1, 2, 3\}$$

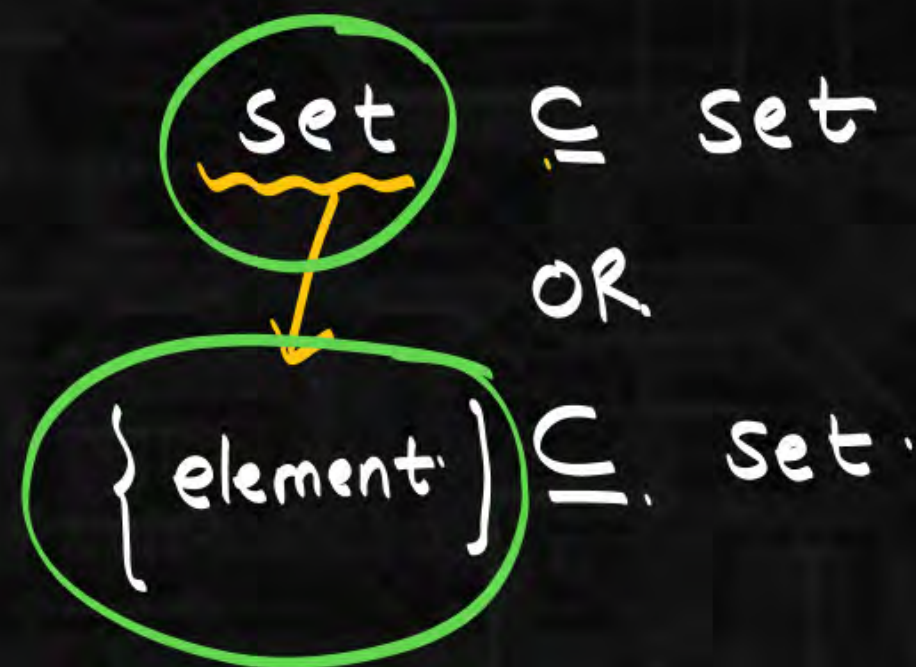
$$P(A) = \{ \overset{e_1}{\cancel{\emptyset}}, \overset{e_2}{\{1\}}, \{2\}, \{3\}, \{23\}, \{13\}, \{12\}, \{123\} \}$$

a	b	c	
0	0	0	$\rightarrow \emptyset$
1	0	0	$\rightarrow \{a\}$
0	1	0	$\rightarrow \{b\}$
0	0	1	$\rightarrow \{c\}$

$\textcircled{a}, \textcircled{b}, \textcircled{c}$

$$2 \times 2 \times 2 = 2^3$$

element $\in$ set



$$x \subseteq A$$

$$\text{set} \subseteq$$

$$\underline{\{ \text{element} \}}$$

$$A = \{ \overset{e_1}{x}, \overset{e_2}{\{x\}}, \overset{e_3}{\{\{x\}\}} \}$$

$$1. x \in A \quad \{x\} \in A \quad 3. \quad (T)$$

$$2. \times x \subseteq A \quad \{x\} \subseteq A \quad 4. \checkmark$$

$$5. \quad \underline{\{\{x\}\}} \in A$$

$$6. \quad \{ \overset{e_2}{\{x\}} \} \subseteq A$$

$$A = \left\{ \frac{1}{e_1}, \frac{\{1\}}{e_2}, \frac{\{2\}}{e_3} \right\}$$

1. $1 \in A$
2. $\{1\} \in A$
3. $\{1\} \subseteq A$
4. $\boxed{2 \in A}$
5. $\{2\} \in A$
6. $\{\{2\}\} \subseteq A$

$$A = \left\{ \frac{\phi}{e_1}, \frac{\{\phi\}}{e_2}, \frac{\{\{\phi\}\}}{e_3} \right\}$$

1. $\phi \in A$ (T) (e_1)
2. $\phi \subseteq A$ (Thm)
3. $\{\phi\} \in A$ (T) (e_2)
4. $\left\{ \frac{\{\phi\}}{e_2} \right\} \subseteq A$
 $\{e_2\} \subseteq$

